2.6 Early Finishers: Trigonometry and Statistics — Markscheme

Version (a)

Extension set, scored separately from the quiz. Total: 17 marks. Both problems are copies of quiz version (b) problems 2 and 4, with the same markscheme. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. Problem 1 needs radian mode.

1. [Maximum mark: 8]

(a) $\dfrac{\sin P\hat{R}Q}{12} = \dfrac{\sin 0.7}{9}$ **M1**

$\sin P\hat{R}Q = 0.858956\ldots$

$P\hat{R}Q = 1.03 \quad (1.03322\ldots)$ **A1**

$P\hat{R}Q = \pi - 1.03322\ldots = 2.11 \quad (2.10836\ldots)$ **A1**

Both values are possible because $0.7 + 2.10836\ldots < \pi$*. Wrong methods: giving only* 1.03 *earns M1A1A0; degree mode (*$\sin P\hat{R}Q = 0.0163$*) earns M1A0A0; answers in degrees (*59.2°*,* 121°*) earn M1A0A0.*

(b) $P\hat{Q}R = \pi - 0.7 - 2.10836\ldots = 0.333 \quad (0.333229\ldots)$ **A1**

Accept 0.332 *from the rounded* $P\hat{R}Q = 2.11$*. Wrong method: using the acute value gives* 1.41*: A0. FT from (a).*

(c) $\dfrac{PR}{\sin 0.333229\ldots} = \dfrac{9}{\sin 0.7}$ **M1**

$PR = 4.57\,\text{cm} \quad (4.56967\ldots)$ **A1**

Accept the cosine rule $9^2 = 12^2 + PR^2 - 2(12)(PR)\cos 0.7$*: it gives* $PR = 4.57$ *or* 13.8*, and the obtuse case is the smaller root; giving* 13.8 *(the acute case) earns M1A0. Accept* 4.55 *from the rounded* $P\hat{R}Q = 2.11$*. FT from (b).*

(d) Area $= \frac{1}{2}(12)(9)\sin 0.333229\ldots$ **M1**

Area $= 17.7\,\text{cm}^2 \quad (17.6631\ldots)$ **A1**

Accept $\frac{1}{2}(12)(PR)\sin 0.7$*. Accept* 17.6 *from the rounded* $P\hat{R}Q = 2.11$*. Wrong method:* $\frac{1}{2}(12)(9)\sin 0.7 = 34.8$ *(0.7 is not between these sides): M0A0. FT from (b) or (c).*

Feedback: in this problem rounding an angle early moves the later answers. Store the full values in the calculator.

2. [Maximum mark: 9]

(a) $6 + p + 12 + q + 3 = 40$, so $p + q = 19$ **A1**

$$\frac{15(6) + 25p + 35(12) + 45q + 55(3)}{40} = 31.25$$ **M1**

$675 + 25p + 45q = 1250$, so $5p + 9q = 115$

$p = 14$ **A1**

$q = 5$ **A1**

M1 for the mean equation using the class midpoints $15, 25, \ldots, 55$*. Accept solving the system on the GDC. Wrong method: lower class boundaries (*$10, 20, \ldots$*) give* $p = 4$*,* $q = 15$*: A1M0A0A0.*

(b) $\sigma = 11.1$ minutes $\quad (11.1102\ldots)$ **M1A1**

M1 for the midpoints with their frequencies entered in the GDC (may be implied by a correct answer). Wrong method: the sample value $s_x = 11.3$ *earns M1A0. FT from (a).*

(c) new time $= 60(t + 2) = 60t + 120$ seconds **(M1)**

new mean $= 60(31.25 + 2) = 1995\,\text{s}$ **A1**

new standard deviation $= 60 \times 11.1102\ldots = 667\,\text{s} \quad (666.614\ldots)$ **A1**

The penalty moves the mean but not the spread; the conversion to seconds multiplies both by 60*. Accept* 666 *from the rounded* $\sigma = 11.1$*. Wrong methods: adding* 120 *to the standard deviation (*787*), or adding the* 2 *minutes after converting (mean* 1877*): A0 for that value. FT from (b).*

La Scuola d'Italia / Dr. Huson / IB Math 12
28 October 2026 Markscheme

Version (b)

Extension set, scored separately from the quiz. Total: 21 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. Problem 1 needs radian mode: a value worked in degree mode earns the M and substitution marks if the method is right, and A0 for the answer.

1. [Maximum mark: 11]

(a) $BD^2 = 10^2 + 10^2 - 2(10)(10)\cos 1.4$ **M1A1**

$BD^2 = 166.006\ldots$

$BD = 12.9\,\text{cm} \quad (12.8843\ldots)$ **A1**

M1 for the cosine rule in triangle ABD, A1 for the correct substitution. Degree mode gives 0.244*: M1A1A0.*

(b) $\cos B\hat{C}D = \dfrac{7^2 + 7^2 - 12.8843\ldots^2}{2(7)(7)}$ **M1A1**

$\cos B\hat{C}D = -0.693944\ldots$

$B\hat{C}D = 2.34 \quad (2.33774\ldots)$ **A1**

M1 for the cosine rule for an angle in triangle BCD, A1 for BD as the side opposite $B\hat{C}D$. The answer must be in radians: $134°$ *earns M1A1A0. Wrong method: a sign error gives* 0.804*: M1A0A0. Accept* 2.34 *from the rounded* $BD = 12.9$*. FT from (a).*

(c) Area $ABD = \frac{1}{2}(10)(10)\sin 1.4 = 49.2724\ldots$

Area $BCD = \frac{1}{2}(7)(7)\sin 2.33774\ldots = 17.6407\ldots$ **M1**

Area $= 49.2724\ldots + 17.6407\ldots$ **M1**

Area $= 66.9\,\text{cm}^2 \quad (66.9131\ldots)$ **A1**

First M1 for a correct area expression for either triangle. Second M1 for adding the two triangles. Accept other correct methods: two copies of triangle ABC, $2 \times \frac{1}{2}(10)(7)\sin A\hat{B}C$, or half the product of the diagonals. Accept any value consistent with the candidate's own correctly rounded intermediates, e.g. 66.8 *from* $BD = 12.9$*. Wrong method: triangle ABD alone,* 49.3*: M1M0A0. FT from (b).*

(d) $2A\hat{B}C = 2\pi - 1.4 - 2.33774\ldots$ **M1**

$A\hat{B}C = 1.27 \quad (1.27271\ldots)$ **A1**

M1 for the angle sum 2π *with $A\hat{B}C = A\hat{D}C$ by symmetry, or for any other valid method (e.g. $A\hat{B}D + D\hat{B}C$). Wrong method: $A\hat{B}D = \frac{\pi - 1.4}{2} = 0.871$ alone: M0A0.* $72.9°$ *earns M1A0. FT from (b).*

2. [Maximum mark: 10]

(a) $\dfrac{12+19+14+k+27+26}{6}=20$ **M1**

$98+k=120, \quad k=22$ **A1**

Wrong method: dividing by 5 *gives* $k=2$*: M0A0.*

(b) $r=0.896 \quad (0.895547\ldots)$ **A1**

$y=1.90x+8.57 \quad (1.90476\ldots, 8.57142\ldots)$ **A1A1**

A1 for a, A1 for b. Accept $a=\frac{40}{21}$, $b=\frac{60}{7}$*. The answer must be an equation; award A1A0 if a and b are given without the equation. FT from (a).*

(c) use the regression line of x on y **M1**

$x=0.421y-2.42 \quad (0.421052\ldots, -2.42105\ldots)$ **A1**

$x=0.421052\ldots(24)-2.42105\ldots=7.68$ hours $\quad (7.68421\ldots)$ **A1**

M1 for choosing the x-on-y line to estimate x from y. Accept 7.68 *from the rounded coefficients. Wrong method: solving the y-on-x line for x gives* 8.10*: M0A0A0. FT from (a).*

(d) $(\bar{x},\ \bar{y})=(6,\ 20)$ **A1A1**

A1 for each coordinate. Both regression lines pass through the mean point $(\bar{x},\ \bar{y})$*;* $\bar{y}=20$ *is given in the question.*